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Department of Artificial Intelligence and Data Science

Academic Year: 2024- 2025 (ODD Semester)

Active Learning Best practices: Think Pair Share -Identity Access Management

Degree, Semester & Branch: V Sem. B.Tech. Artificial Intelligence and Data Science

Course Code & Title: - CCS335 – Cloud Computing

Name of the Faculty member: Mr. P. VETRIVEL, AP/AI & DS

Theme of discussion: Think Pair Share

Topics Covered: Unit - V

Date and Time: 14.11.2024 & 05.00 P.M to 06.00 P.M

Active Learning Best practices: Think Pair Share – Identity Access Management

Unit / Topic: Unit – V / Identity Access Management

Course Outcome: CO5 - Explain security challenges in the cloud environment.

Topic Learning Outcome: TLO31

Activity Chosen: Think Pair Share

Justification:

As organizations increasingly adopt cloud services, managing identities and access becomes crucial for maintaining security, compliance, and operational efficiency. Here are the key justifications for implementing IDM in cloud environments:

- Enhanced Security
- Centralized Identity Management
- Scalability and Flexibility

Time Allotted for the Activity: 30 Minutes

Details of the Implementation:

Identity Management (IDM) is a framework for managing and controlling user identities and their access to various resources within an organization. It ensures that only authorized individuals can access specific systems and data.

Key Components of IDM:

1. **User Identity Creation**: Establishes unique identities for users, typically with usernames and other attributes (e.g., role, department).
2. **Authentication**: Verifies the identity of users through methods such as passwords, multi-factor authentication (MFA), or biometrics.
3. **Authorization**: Grants or restricts access to systems based on user roles, permissions, and policies.
4. **User Provisioning and De-provisioning**: Manages the creation, modification, and deletion of user accounts, ensuring timely access changes when users join or leave the organization.
5. **Single Sign-On (SSO)**: Allows users to authenticate once and gain access to multiple systems without re-entering credentials.
6. **Access Control**: Defines who can access what resources and at what level (e.g., read, write, admin).
7. **Audit and Monitoring**: Tracks user activity and access to ensure compliance, detect anomalies, and provide a record for audits.

Correctness

The correctness of Identity Management (IDM) refers to the accuracy, reliability, and appropriateness of how identities are created, managed, and authenticated in a system. Ensuring correctness in IDM is crucial for maintaining security, preventing unauthorized access, and ensuring that users have the appropriate level of access to resources. Here are key factors that determine the correctness of IDM:

- Accurate User Identity Creation and Management
- Accurate Authentication
- Correct Access Control and Authorization
- Timely Provisioning and De-provisioning
- Audit and Monitoring
- Consistency with Policies

CO – PO / PSO Mapping

CO	PO1	PO2	PO5	PO9	PO10	PO11	PO12	PSO3
CO5	3	3	2	2	2	2	2	3

PO/PSO Mapped

PO1: Students can be able to apply the principles of computer science and engineering to solve a security challenge in cloud environment.

PO2: Students can be able to compare and contrast alternative solution/methods for various security challenges.

PO5: Students can be able to identify modern engineering tools, techniques and resources to prevent the cloud environment from security challenges.

PO9: Students can be able to demonstrate effective communication, problem-solving skills by explain the security challenges in cloud environment.

PO10: Students can be able to read, understand and interpret technical and non-technical information of security challenges in cloud environment.

PO11: Students can be able to identify the tasks required to overcome the security challenges in cloud environment.

PO12: Students can be able to recognize the need and be able to clearly explain security challenges to build secure cloud environment.

PSO3: Students will be able to enrich the critical thinking skills in cloud technologies with mathematical aid to foresee the research findings and provide the solutions.

Glimpses:

Reflective Critique:

- Preparing and Sharing Content: Video lectures and reading materials were created covering the relevant topics. These materials were shared with the students through the learning management system or other suitable platforms well in advance of the corresponding class sessions.
- Class Sessions: During the class sessions, students were asked to discuss and apply the concepts covered in the pre-recorded lectures. Facilitators were present to guide and support the students' learning process.
- Active Learning Activities: Students are asked to explain about the concept in front of the classroom, and peer teaching sessions were conducted.
- Feedback Collection: Students were asked to provide feedback on the flipped classroom experience, highlighting the aspects they found beneficial and suggesting potential improvements.

Observations:

The implementation of the Think Pair Share was successful in enhancing student engagement, promoting deeper understanding, and creating a more interactive learning environment. By using this approach, students were encouraged to take ownership of their learning process and actively participate in class activities. However, continuous refinement based on student feedback and ongoing assessment of the approach will be essential to optimize its effectiveness in future implementations.

Benefit of the practice:

- The students understand well about IDM.
- The students can able to Interact with the faculty member and their class mates during the class was effective.
- It helps the students to improve the self-learning of a student.
- With the help of this activity, each student can get clear idea about the use of snapshot algorithms in distributed systems.

Students Response:

- Bright students were actively participated to prepare the topic wise questions and concept notes.
- Slow learners expect to assist the topic towards searching online and also understanding the questions.
- Students' response ensures the students could improve listening, communication, creativityand problem-solving skills.

Signature of the Faculty Member **HoD**

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